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Photocatalysis

Excitonic Effects in Polymeric Photocatalysts

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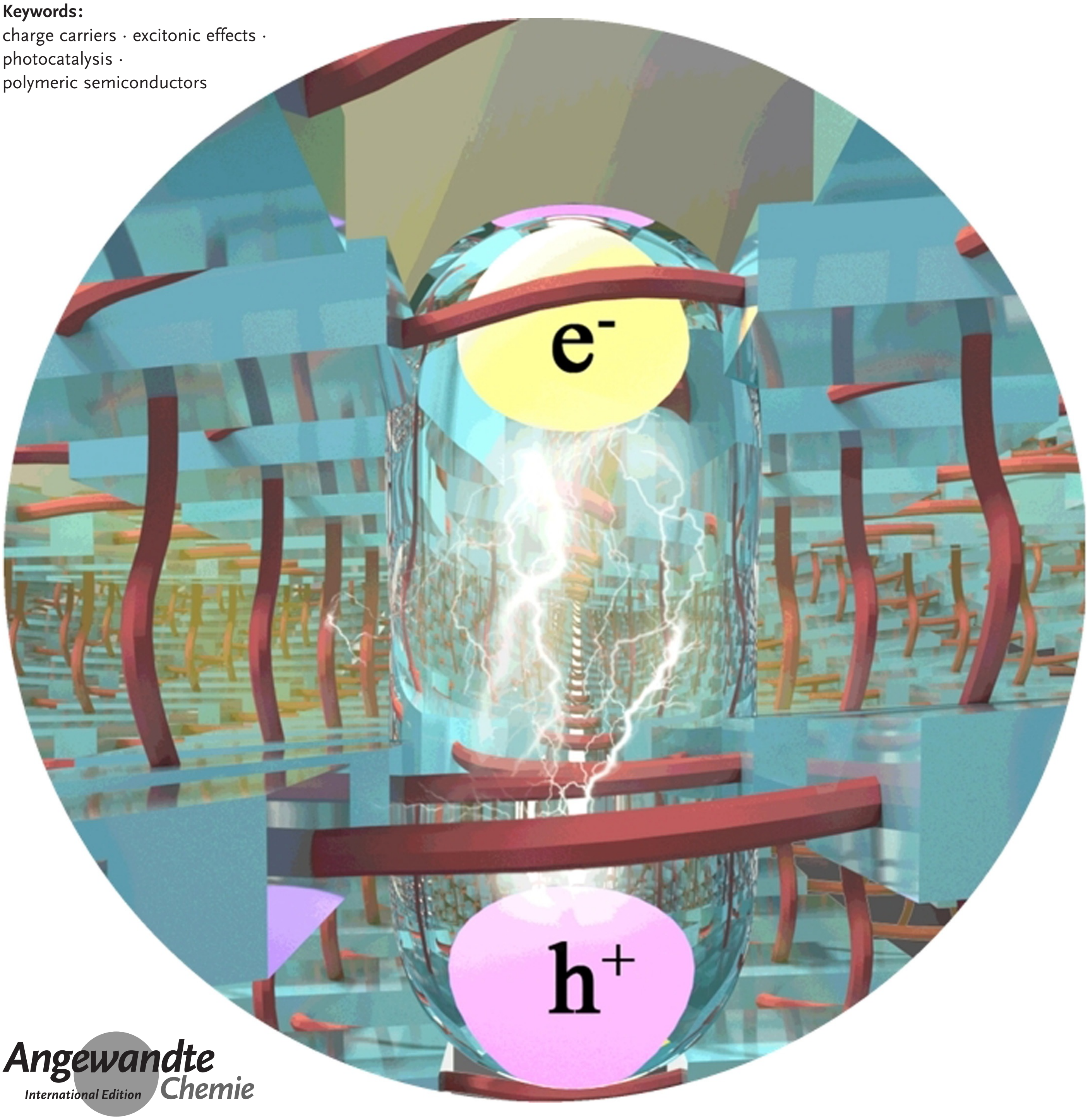

Owing to the intrinsically low dielectric properties, robust Coulomb interactions between photoinduced electrons and holes lead to dramatically strong exciton effects in polymeric photocatalysts. Such effects endow polymeric matrixes with nontrivial photoexcitation processes determining photocatalytic energy utilization. In this Minireview, we describe recent progress in the investigation of the excitonic effect in polymeric photocatalysts. On the basis of the understanding of excitonic effects in polymeric systems, we outline the relationships between excitonic behaviors and photocatalytic performance. Advances in optimizing the excitonic effect for gaining high-efficiency polymer-based photocatalysis are summarized. We also discuss the challenges in the field and forecast the directions for future research.

1. Introduction

Over the past decades, photocatalytic research has been attracting tremendous attention by virtue of its great potential in realizing large-scale conversion of solar energy to chemical energy.[1–5] As the platforms for photocatalysis, catalysts characterized by desirable electronic and optical properties are crucial in pursuing photocatalytic energy utilization. With great efforts devoted to designing advanced photocatalysts, various classes of materials have been established, among which polymeric semiconductors have been highlighted by virtue of the general advantages like low cost, simple synthetic procedure, and weak biotoxicity.[5–10] As a typical kind of macromolecules, polymers consist of thousands of repeating molecular subunits (namely, monomers), whose electronic structures are closely related to the factors like monomer species, aggregation states, polymerization degrees. In general, constituent monomers with conjugated features are generally favorable, where the π-electron delocalization endows the systems with suitable visible-light response and excellent charge carrier transport.[11,12] On the basis of this convention, numerous conjugated monomers such as tri-*s*-triazine/*s*-triazine,[4] pyrene,[13] benzothiadiazole,[14] and imide[15] have been proposed for constituting polymeric photocatalysts. There are also polymeric photocatalysts comprised of two or more species of monomers through cross-coupling.[16] Benefiting from alternate arrangement of different species, these co-polymers are deemed to be effective in charge separation which facilitates their performance in charge-carrier-involved photocatalytic reactions. Recently, a series of low molecular weight oligomers has also been proposed for photocatalytic reactions.[17–19] It is noteworthy that most of the above mentioned polymers are amorphous, due to the lack of long-range order. The amorphous feature, on the one hand, renders rich reactive sites and promoted charge separation, on the other hand, impede the evaluation of the influences of specific structures on photocatalytic behaviors. Recently, the emerging polymer frameworks (i.e., covalent organic frameworks) present supplements to the amorphous polymeric photocatalysts, where the extraordinary crystalline and porous structures in the frameworks result in quite non-trivial catalytic behaviors.[20] As compared with other photocatalysts including complexes, noble metals, and inorganic semiconductors, polymeric semiconductors possess much more abundant structural configurations such as varied monomer compositions, flexible polymer chains, spatial inhomogeneities. The structural diversities lead to not only remarkably abundant electronic and optical properties but also rich possibilities in controllable regulation.

In pursuit of advanced polymeric photocatalysts, numerous attempts at understanding and optimizing the involved photoexcitation processes have been made.[5,8–10] A habitual charge-carrier concept is typically employed to understand photoexcitation processes of polymeric photocatalysts, where the behaviors of non-equilibrium charge carriers are mainly interrogated. On the basis of this understanding, extensive investigations have been made in optimizing the properties including optical absorption, band-edge potentials, charge transport, and photoinduced carrier relaxation.[21–25] Nevertheless, there are some photocatalytic phenomena that can hardly be explained. Taking polymeric carbon nitride as an example, photocatalytic singlet oxygen ($^1O_2$) generation are sometimes efficient in the system, which seem to be independent of superoxide anion radical generation;[26] the intrinsic photocatalytic performances of bare polymeric systems tend to be quite insufficient, and heterostructures are usually involved in achieving high-efficiency photocatalysis;[5] the quantum yields of photocatalysis and photoluminescence, which are generally considered to be opposite with each other, are both extremely high in two-dimensional polymeric carbon nitride.[27,28] The reason for these phenomena is that traditional viewpoints focusing on charge-carrier photophysical properties seem to simplify the involved photoexcitation processes, where the long-ignored excitonic effects in polymeric catalysts lead to much more complicated but interesting photoexcitation properties.

Excitonic effects, which are mediated by Coulomb interactions between electrons and holes, play a significant part in the optical and electronic properties of semiconductors.[29,30] Nevertheless, the potential influences of excitonic effects have seldom been considered in respect of semiconductor-based photocatalysis. One of the primary reasons is

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that, as for the extensively investigated inorganic semiconductor-based photocatalysts, high dielectric properties (dielectric constants of 10–15) give rise to faint electron-hole interactions and hence small exciton binding energies. Low exciton stabilities can be expected in these systems once exciton binding energies are lower than 26 meV (that is, the thermal energy under room temperature, $k_B T$). As such, excitonic processes are substantially covert as compared to the charge-carrier counterparts. Situations can be quite distinct in polymeric photocatalysts: in view of the intrinsically low dielectric properties (dielectric constants of 3–5), polymeric semiconductors tend to possess much larger exciton binding energies, and hence excitons or bound electron-hole pairs, rather than free carriers, will dominate the photoexcitation processes therein.[31,32] Moreover, nonlinear photoexcitation behaviors stemming from Coulomb/dipole interactions between excitons further complicate photocatalytic discussions in polymeric systems. For instance, the correlations between excitons lead to robust exciton-exciton annihilation in polymers, which sets limitations to the overall photocatalytic efficiency.[33] With respect to polymeric photocatalysts, the long-ignored excitonic aspect of photoexcitation processes bottlenecks the understanding of the relevant photocatalytic mechanisms and the optimization of photocatalytic energy utilization.

Recently, a series of investigations has demonstrated that the excitonic aspect, which serves as a supplement to the traditional charge-carrier aspect, should be widely considered in dealing with polymer-based photocatalysis. However, the systematically understanding of the influences of excitonic effects on polymeric photocatalysis remains lacking. In this Minireview, we focus on the excitonic and charge-carrier aspects of the photoexcitation processes in polymeric photocatalysts, where the relationships between the two aspects are highlighted. Further, the influences of excitonic effects on photocatalytic behaviors of polymeric semiconductors are systematically summarized. On the basis of the understanding of excitonic effects-dominated mechanisms and efficiencies of energy-utilization, we describe recent progress in excitonic regulation for achieving high-efficiency polymer-based photocatalysis, where the relationships between structural factors and excitonic processes are emphasized. Finally, we forecast the challenges and prospects in the excitonic aspect in polymeric photocatalysts.

2. Charge-Carrier and Excitonic Aspects in Photoexcited Polymeric Systems

In terms of polymer-based photocatalysis, traditional viewpoints mainly focus on charge-carrier behaviors, where the classical band model is generally employed to describe the charge-carrier-related photoexcitation processes.[2,5] As depicted in Figure 1 a, for gaining effective photoexcitation, the absorption of photons with suitable energy (that is, larger than band gaps) is required. Upon excitation, hot electrons and holes can be formed in conduction and valance bands, followed by ultrafast intra-band relaxations toward the corresponding band edges. Band-edge charge carriers decay undergoing non-radiative or radiative or some other (e.g., defect-trapping) pathways. The migration of these band-edge charge carriers to the surface is necessary for photocatalysis. The transfer of electrons/holes from polymer-based photocatalysts to the absorbed substrates leads to the formation of reduction/oxidation products, during which the match between energy levels of band edges and redox potentials of half reactions are required. In the context of the charge-carrier viewpoint, several fundamental factors, including band-edge

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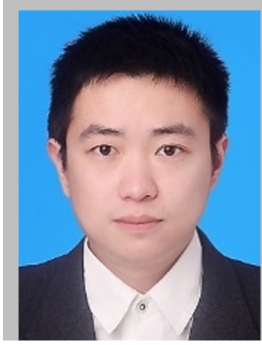

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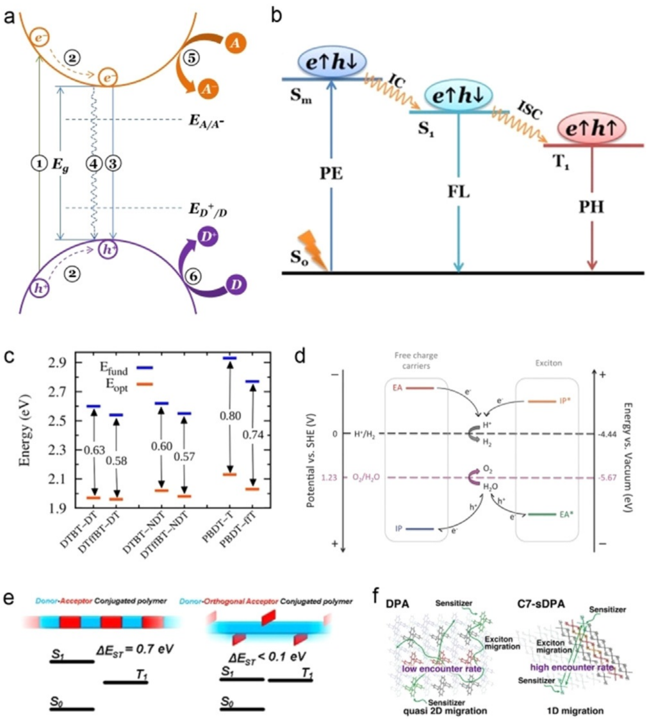

Figure 1. (a) Schematic illustration of the charge-carrier aspect in polymeric photocatalysts, where the processes of 1–6 denote photo-excitation, intra-band relaxation, radiative decay, non-radiative decay, photo-reduction, and photo-oxidation, respectively. (b) Schematic illumination of the excitonic aspect of photoexcitation processes in polymers, where PE, FL, PH, IC, and ISC denotes photoexcitation, fluorescence, phosphorescence, internal conversion, and intersystem crossing, respectively. (c) Calculated fundamental and optical gaps of polymer pairs without and with fluorine, using polarizable continuum model. (d) Schematic illustration of the influences of excitonic effects on band-edge potentials of polymeric photocatalysts. (e) Schemed the structures and singlet-triplet energy gaps of donor-acceptor- and donor-orthogonal acceptor-based conjugated polymers. (f) Schematic illumination of the effects of molecular orientation on triplet-triplet annihilation. (c) Adapted with permission from Ref. [40] Copyright 2019 American Chemical Society. (d) Adapted with permission from Ref. [43] Copyright 2016 Wiley. (e) Adapted with permission from Ref. [47] Copyright 2017 American Chemical Society. (f) Adapted with permission from Ref. [55] Copyright 2018 American Chemical Society.

potentials, charge-carrier recombination and separation, electronic transport, and reactive site, are crucial. Nevertheless, some additional, distinct photoexcitation processes can be expected in polymeric systems: in contrast to the strong dielectric properties of inorganic semiconductors, the intrinsically low dielectric properties of polymeric systems would lead to robust correlations between non-equilibrium charge carriers. These correlations will lead to large exciton binding energies (usually hundreds of meV) and hence considerable generation of excitons or bound electron-hole pairs, which can greatly influence the electronic and optical properties of polymers. Taking polymeric carbon nitride as an example, theoretical simulations reveal that the nonionicity and low polarizability of the metal-free framework lead to a low dielectric property, which gives rise to the formation of highly localized excitons with large binding energy in tri-*s*-triazine units.[34–36] Besides, experimental investigations demonstrate that the photoluminescent properties of carbon nitride polymers can be described by the excitation and radiative recombination of molecular excitons confined in tri-*s*-triazine units, which suggests that, in terms of photophysical processes, polymeric carbon nitride tends to behave like a monomer, rather than an inorganic semiconductor.[37] In this context, the excitonic and charge-carrier aspects would collectively make up the photoexcitation processes of polymeric systems. Moreover, considering that excitons are bound states of electrons and holes mediated by Coulomb interactions, the excitonic and charge-carrier aspects of photoexcitation processes in photoexcited polymers are in fact relevant, rather than independent. In this case, clarifying the relationships between the two aspects would be a meaningful subject, not only for understanding the photoexcitation processes, but also for designing advanced polymeric photocatalysts.

Traditionally, Jablonski model is employed to describe the excitonic aspect in photoexcited polymeric systems (Figure 1b), where the processes including excitation, emission, and relaxation are quite different from charge-carrier ones. One of the primary principles in processing relationships between the excitonic and charge-carrier aspects is to consider their different energies. The energy-level difference between the conduction and valance band edges is termed energy gap, of which the value is equal to the energy of a free (that is, without correlations) electron-hole pair. By contrast, reduced energy can be expected in the corresponding exciton or bound electron-hole pair, due to the partial energy release during the Coulomb interaction-correlation between electron and hole. The value of energy release is so-called exciton binding energy, which can be used to estimate the strength of excitonic effects in semiconductors. As for polymers, exciton binding energy can be effectively regulated by structural modifications.[38–40] For instance, Benatto et al. theoretically investigated the crucial role of fluorine substitution on regulating of electronic structures in polymers, where the reduced exciton binding energy through fluorine substitution were highlighted, owing to the promoted molecular planarity-induced electronic delocalization (Figure 1c).[40] It is worth noting that the optical band gaps estimated by absorption spectra are virtual the excitonic gaps of polymers. That is to say, as for polymeric systems, optical band gaps estimated from ultraviolet-visible spectra are actually the energy of primary excitons.[36,40] The energy gap between valance and conduction bands in polymeric case can be estimated by photoconductivity spectroscopy,[32] where the threshold denotes single-particle band gap or fundamental gap, corresponding to the lowest energy for generating free electrons and holes. Besides, the energy difference between excitons and free electron-hole pairs enables the respective tracking of the excitonic and charge-carrier aspects to be made by spectroscopic investigations. For instance, the exciton and charge-carrier relaxation dynamics in polymeric carbon nitride have recently been tracked by means of transient absorption spectroscopy, taking advantage of the distinct energy profiles of the two aspects in the system.[33,41,42] Band model of molecular systems would provide a comprehensive understanding of exciton-involved band structures of polymeric photocatalysts (Figure 1 d):[43] regardless of excitonic effects, ionization potential (IP) and electron affinity (EA) corresponds to the energies of HOMO and LUMO, their energy gap is referred to as fundamental gap; once the correlations between electrons and holes are considered, EA* and IP* can be used to denote the potentials of a polymer with a localized exciton to obtain and loss an electron. The energy gap between EA and IP* (or between EA* and IP) is exciton binding energy. For gaining feasible photocatalytic reactions, the potentials of IP/EA* and EA/IP* should match with redox potentials of reduction and oxidation reactions.

Besides, the charge conditions and spin configurations of free carriers and excitons should be considered in dealing with the photoexcitation processes of polymeric photocatalysts. As compared to the positively/negatively charged holes/electrons, excitons are electrically neutral quasi-particles. The distinct charge conditions between carriers and excitons lead to different responses toward electric field, thereby giving rise to different mechanisms in photophysical processes like migration, relaxation, and decay. For instance, photoinduced built-in electric fields in semiconductors are believed to promote the spatial separation of free electrons and holes,[44] which, however, tend to be ineffective to dissociate excitons into free charges.[45] Another feature is the different spin configurations. As compared to the half-integral spin of individual charge carriers, excitons tend to carry a total spin of either zero or one, corresponding to spin-singlet or -triplet configuration, respectively.[46] Owing to spin conservation, singlet and triplet excitons possess distinct photophysical properties like lifetime and energy. For instance, the radiative decays of singlet and triplet excitons result in nanosecond-scale fluorescence and microsecond-scale phosphorescence, respectively, which enable the selective extractions of the two classes of excitons in spectroscopic investigations. The intriguing spin relaxation dynamics of excitons in polymers occupies an important place in photoresponsive applications, and inevitably influences the corresponding photocatalytic performance. For the spin-relaxation processes in polymers, Bronstein et al. reported that the intersystem crossing for relaxing singlet excitons to triplet ones can be effectively optimized by structural construction (Figure 1 e).[47] They demonstrated that the incorporation of orthogonal acceptor can spatially separate electron and hole wave functions, which give rise to greatly reduce the singlet-triplet energy gap.

Given the strong interactions between photoinduced species (that is, including charge carriers and excitons), high-order correlations can also be expected in polymeric systems, which endow the systems with some unique photophysical properties. For instance, exciton-exciton annihilation arising from the correlations between excitons represents a nonlinear, high-order recombination process in polymers.[48–50] That is, the collision between two low-lying excitons leads to the elimination of one exciton, while the residual one is excited to a high-lying state. In fact, with regard to the charge-carrier aspect, there is a similar a non-radiative process, Auger recombination involving the energy re-distribution in a three-carrier system have been widely investigated in pursing nonlinear optical properties in photoresponsive materials.[51,52] Therefore, it is rational to expect the great potential of high-order correlations between excitons in polymer-based photocatalytic applications. Besides, the high-order correlations between excitons can be affected by the concentration of excitons. This feature implies that the high-order correlations in polymers can be regulated by structural engineering,[53,54] which promises the great potential in designing advanced photoresponsive polymeric materials. For instance, Sato et al. theoretically investigated that triplet-triplet annihilation in aggregates can be regulated by molecular orientation strategies, where the directional triplet exciton migration greatly promote the encounters of excitons (Figure 1 f).[55] It is noteworthy that high-order correlations are highly dependent on excitation scenario, where high exciton density is favorable. However, in terms of photocatalysis, the exciton-density thresholds of high-order correlations in polymeric photocatalysts have not yet been fully investigated. Researches on other photoresponsive polymeric materials have demonstrated the corresponding thresholds in different polymers deviated greatly from each other.[54,56,57] Besides, it can be concluded that high-order correlations in polymeric systems are closely associated with the properties like exciton diffusion and dipole-dipole couplings, which means that these correlations could be optimized by structural modifications such as aggregate designing and polymerization-degree control.[58,59]

3. The Excitonic Aspect for Polymer-based Photocatalytic Energy Utilization

In polymeric systems, excitonic effects dominate the electronic and optical properties of the matrixes, and hence lead to inevitable influences on the relevant photocatalytic behaviors. Due to the differences in properties like energy level, spin-configuration, and charge condition, the excitonic and charge-carrier aspects tend to possess different contributions toward photocatalysis.

The excitonic aspect in polymeric systems establishes an alternative pathway for realizing photocatalytic energy utilization. As for traditional investigations on polymeric photocatalysis, carrier-based charge transfer process between catalyst and substrate is responsible for the photocatalytic behaviors in the system. For instance, in photocatalytic oxidation or reduction reactions, photogenerated electron or hole with certain potential energies in catalyst transfer toward substrate molecule, leading to the formation of reductive or oxidative species in the system. Given that the process involves the net transfer of photoinduced charge carrier between catalyst and substrate, two necessary conditions should be met: the contact between the two components and the matched redox and band-edge potentials. Accordingly, the engineering of surface and energy band structures and electronic properties has been extensively accepted to be favorable to photocatalytic performance. By comparison, the excitonic aspect brings about a distinctly different energy utilization pathway. In detail, exciton-based energy transfer process represents a feasible mechanism for photocatalytic energy utilization beyond carrier-based charge transfer process, where energy exchange between excited exciton in catalysts and ground-state substrate are mediated by dipole-dipole interaction or electron exchange interaction. This process is typically called resonant energy transfer, embodying the matched energy level and spin configuration between exciton-donor catalyst and acceptor substrate. For instance, $^1O_2$ generation can be more appropriately associated to exciton-based energy transfer between ground-state oxygen molecules and triplet excitons in polymeric photocatalysts (Figure 2a). The co-existence of the excitonic and charge-carrier aspects in polymeric photocatalysts enable the selectivity of reaction to be controlled by engineering the involved photoexcitation processes.[26,60] Aside from the influence on the mechanism of photocatalytic energy conversion, exciton-based energy transfer process also dominates the other processes related to energy redistribution in polymers. For instance, exciton diffusion typically undergoes exciton-based energy transfer process, where long-range (≈ 10 nm) dipole-dipole interaction leads to the hopping of exciton between different building blocks (Figure 2b).[61] By contrast, the migration of charge carriers is mainly dominated by short-range (less than 1 nm) scattering route. This feature suggests that controllable exciton diffusion can be realized by rationally designing the building blocks of polymer matrix,[59,62] which would be favorable to energy-transfer-mediated photocatalytic energy utilization.

The excitonic aspect also sets limitations to the overall photocatalytic efficiency of polymeric systems. First, given that exciton consists of an electron and a hole, the photoinduced generations of exciton and charge-carrier in polymeric photocatalysts tend to be opposite and complementary. Generally, a remarkable excitonic aspect can be expected in polymers with large exciton binding energy, while a prominent charge-carrier aspect usually emerges in polymers with small exciton binding energy. It is noteworthy that some photoexcitation processes including exciton dissociation and exciton/charge-carrier trapping can also influence the balance between the excitonic and charge-carrier aspects in the systems, which pave the way to the regulation of the photocatalytic behaviors. Another factor influencing energy utilization efficiency is the presence of high-order correlations in polymers. Taking exciton-exciton annihilation as an example, the inelastic collision between two low-lying excitons into one hot exciton (Figure 2c, pathway II) reveals the depopulation of photoinduced species within several hundred picoseconds or several nanoseconds. On the one hand, considerable energy dissipation can be expected due to the rapid hot-exciton cooling process (pathway III), which is undoubtedly detrimental to the overall photocatalytic energy utilization efficiencies;[33] on the other hand, the hot exciton generation via exciton-exciton annihilation implies the potential of realizing low-energy light-driven photocatalysis in polymers (pathway IV). Besides, the excitonic aspect also influences the optical absorption properties of polymers. Different from the optical absorptions of inorganic semiconductors originating from band-to-band transitions, the optical absorption of polymers is dominated by excitonic transitions. However, owing to the factors likes weak spin-orbit coupling and high conjugation degrees, excitonic absorption coefficients are somehow low in some polymeric semiconductors, which clearly limit their spectral response in photocatalytic applications. In this case, optimizing the excitonic absorption of polymeric semiconductors will be beneficial for maximizing solar light harvesting.

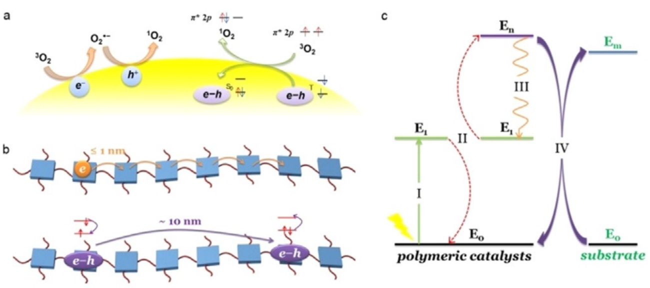

Figure 2. Schematic illustrations of the charge-transfer- and energy-transfer-mediated (a) $^1O_2$ generation and (b) charge/exciton transports in polymeric photocatalysts. (c) Schematic illustration of the effects of exciton-exciton annihilation on polymer-based photocatalysis, where the photophysical processes of I, II, III, and IV are photoexcitation, exciton-exciton annihilation, hot-exciton cooling, and hot-exciton-based energy transfer between polymers, respectively.

A key conception should be kept in mind: the excitonic and charge-carrier aspects collectively constitute the photoexcitation processes of polymeric photocatalysts. Owing to the thermodynamic equilibrium between excitons and charge carriers, any attempt to regulate photoexcitation processes would lead to the simultaneous change of the two aspects. For instance, charge-carrier scavengers have been widely used in photocatalytic reactions, which can also influence the excitonic aspect in polymeric photocatalysts. First, reduced exciton concentration can be expected. Once electron or hole is selectively estimated, the equilibrium is expected to move toward the charge-carrier side, thereby resulting in reduced exciton concentration. Second, the addition of scavenger results in the reduction of excitonic lifetime. The equilibrium movement from exciton to charge carrier opens up an important non-radiative excitonic relaxation pathway, where accelerated exciton depopulation results in reduced excitonic lifetime. Besides, the abundant interactions between molecular scavengers and photocatalysts might lead to complicated excitonic energy redistribution in photocatalytic reactions. In this respect, theoretical investigation revealed that the oxidation of methanol could undergo an excitonic interfacial proton-coupled electron transfer mechanism, where the dissociation of O–H bond is associated with electron-hole separation.[63]

According to the above discussions, the influences of the excitonic aspect on polymer-based photocatalysis are highlighted. The excitonic and charge-carrier aspects play different but complementary roles in photocatalytic applications. The intriguing relationships between the two aspects imply that high-efficiency photocatalytic energy utilization could be achieved by the global optimization of the excitonic and charge-carrier processes.

4. Optimizing the Excitonic Properties in Polymeric Photocatalysts

Given that excitonic properties of polymers including excitation, diffusion, recombination, and annihilation are remarkably associated with structures, it is reasonable to achieve excitonic regulation via structural engineering. Benefiting from the abundant structural diversity of polymers, a series of efforts has recently been devoted to regulating the excitonic aspect of polymeric photocatalysts, which not only gave rise to optimized strategies on photocatalytic energy utilization, but also established understandings of the involved photoexcitation processes. In this section, we review the recent progress in the optimization of the excitonic properties in polymeric photocatalysts.

For charge-transfer-mediated photocatalytic reactions, photoinduced electrons and holes are the primary reactive species that are responsible for the oxidation or reduction of substrates. However, as for polymeric semiconductors, the insufficient charge carrier generation due to the intrinsically large exciton binding energies limits the relevant photocatalytic performances.[43] In view of the components of an exciton (that is, an electron and a hole bound by Coulomb interaction), one can expect promoted charge-transfer-mediated photocatalysis to be achieved by dissociating excitons into charge carriers. As is well-known, exciton dissociation usually occurs in the areas containing energetic inhomogeneity.[64,65] In this case, introducing structural factors with disordered energy landscape would facilitate those charge-transfer-mediated photocatalysis. Recently, by taking heptazine-based melon as a prototype, we demonstrated that the incorporation of order-disorder interfaces can greatly improve exciton dissociation in the system (Figure 3a).[60] Theoretical simulations and electrochemical measurements suggested that free electrons and holes induced by exciton dissociation can separately inject to the ordered and disordered regions, which further suppresses the re-bound of electrons and holes into excitons. Benefiting from this feature, semicrystalline heptazine-based melon with abundant order-disorder interfaces exhibits significantly reduced exciton accumulation, as revealed by photoluminescence measurements (Figure 3b). As such, semicrystalline heptazine-based melon exhibits significantly enhanced performance in charge-transfer-mediated photocatalytic reactions like superoxide anion radical generation and selective benzyl alcohols oxidation. Also in heptazine-based melon, Wang et al. reported the design of a bioinspired donor-acceptor system for promoting exciton dissociation (Figure 3c).[66] They fabricated a prototypical donor-acceptor system by functionalizing heptazine-based melon with selenium atoms and cyanamide groups. Selenium and cyanamide groups serve as electron donor and acceptor, respectively. Benefiting from the homojunction-like donor-acceptor motifs, the modified heptazine-based melon exhibits enhanced exciton dissociation, thereby giving rise to high-efficiency carrier-based photocatalytic hydrogen generation. The interlayer stacking regulation has also been demonstrated to facilitate exciton dissociation in polymer-based photocatalysts.[67] Wang and co-workers reported the construction of conjugated donor-acceptor polymer photo-

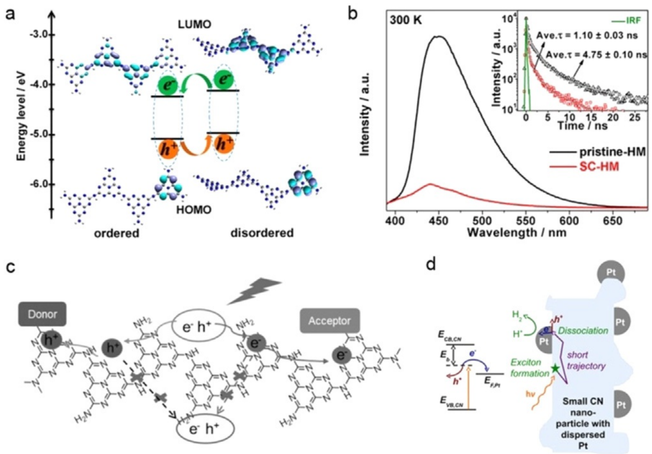

Figure 3. (a) Simulated electronic structures of ordered and disordered heptazine-based chains. Inset: the resulting electron and hole transfer between the two chains. (b) Room-temperature steady-state and time-resolved photoluminescence spectra of pristine and semicrystalline heptazine-based melons. (c) Schematic illustration of exciton dissociation in heptazine-based melon containing donor and acceptor groups. (d) Illustration of noble metal nanoparticle-induced free charge-carrier generation in polymeric carbon nitride structures. (a–b) Adapted with permission from Ref. [60] Copyright 2017 American Chemical Society. (c) Adapted with permission from Ref. [66] Copyright 2018 Wiley. (d) Adapted with permission from Ref. [35] Copyright 2015 American Chemical Society.

catalysts with electron-output "tentacles", where the incorporated dibenzothiophene-*S,S*-dioxide units serve as reactive sites to enhance exciton dissociation.[68] The reduction of exciton binding energy can also promote exciton dissociation in the system, which have been recently achieved by constructing donor-acceptor-based structure.[69] In addition, extrinsic heterostructures (such as, the mostly well-known noble metal-based co-catalysts) are usually employed, where the contained interfaces serve as reactive areas for boosting exciton dissociation into free charge carriers (Figure 3d).[35] Kosco et al. reported the construction of heterojunction nanoparticles for boosting the performance of photocatalytic hydrogen generation, where they highlighted the importance of donor/acceptor heterostructures in facilitating exciton dissociation.[70] Qiao and co-workers have demonstrated van der Waals heterojunction between metal-free phosphorene/g-C_3N_4 nanosheets can effectively promote exciton dissociation, thus giving rise to excellent performance of photocatalytic hydrogen generation under visible light illumination.[71] Surface properties are crucial to the photocatalytic behaviors of polymers.[72,73] Guiglion et al. carried out theoretical simulations to investigate exciton dissociation during photocatalytic reactions, where the importance of the polymer-solution interface was highlighted.[43] Cooper and co-workers investigated the structure–activity relationships in linear polymer photocatalysts, where they demonstrated that the incorporation of sulfone groups into linear hydrophobic backbone can promote H_2O localization; the promotion can facilitate exciton dissociation and hence boost photocatalytic hydrogen evolution.[74] They also investigated the influences of excitonic effects in sulfone-containing covalent organic frameworks, where the exciton-dominated absorption properties and high hydrophily-induced exciton dissociation were demonstrated.[75]

The regulation of exciton spin relaxation represents another intriguing issue in polymeric photocatalysts. The harvesting of excitons with certain spin property is quite necessary for resonant energy transfer. For instance, for gaining efficient photocatalytic $^1O_2$ generation, triplet exciton harvesting in catalyst is required. In polymeric photocatalysts, long-lived triplet excitons are generated through the relaxation of singlet excitons, undergoing intersystem crossing to overcome spin forbidden between the two states (Figure 4a). Nevertheless, owing to the typically weak spin-orbit coupling and large singlet-triplet energy gap, the intrinsic spin relaxation in pristine polymers is insufficient, leading to an insufficient triplet exciton generation. In this case, searching effective strategies to promote spin relaxation dynamics in polymeric semiconductors is needed. To this issue, we have demonstrated the functionalization of polymeric carbon nitride with carbonyl groups can regulate the spin relaxation dynamics in the system.[26] According to spectroscopic investigations, we confirmed that the incorporation of carbonyl groups lead to the synergistic optimizations of spin-orbit coupling and singlet-triplet energy gap (Figure 4b) of the matrix, which leads to a remarkable promotion in intersystem crossing efficiency. Benefiting from this, the oxidized carbon nitride with abundant carbonyl groups exhibits considerable triplet harvesting, thereby showing excellent photocatalytic $^1O_2$ generation (Figure 4c). The addition of heavy atoms can also substantially improve spin-orbit coupling,[33,76] which has been widely employed for regulating spin relaxation dynamics in polymeric photoresponsive applications. For instance, by combining photoluminescence spectroscopy and ultrafast transient absorption spectroscopy (Figure 4d,e), we have recently demonstrated that the addition of 2-bromoethanol can significantly promote spin-orbit coupling of polymeric carbon nitride, hence giving rise to an enhanced intersystem crossing rate.[33]

As discussed above, high-order correlations between excitons are closely related to the efficiency of photocatalytic energy utilization in polymeric semiconductors, whereas their significance has rarely been noted in respect of photocatalytic studies. Besides, in view of its contradictory effects on the sides of photoinduced species depopulation and photoactive wavelength region in polymers, the regulation of high-order correlations tends to be attractive but tricky. In this respect, there are few available typical cases breaking this new ground. For instance, focusing on the abnormal high efficiencies of both photocatalytic and photoluminescence behaviors in two-dimensional polymeric carbon nitride, we highlighted that the remarkably suppressed triplet-triplet annihilation in the matrix is the main factor.[33] Benefiting from the promoted conjugated feature, the promoted singlet-triplet energy gap and reduced spin-orbit coupling can be expected in two-dimensional polymeric carbon nitride, thus leading to substantially reduced intersystem crossing (Figure 5a–d). The resulting faint triplet exciton generation hence gives rise to suppressed triplet-triplet annihilation in the system. The other side of high-order correlations establishes an alternative pathway of extending the light absorption.[77–79] For instance, Kim et al. reported the achievement of sub-band-gap photocatalysis of platinum-loaded tungsten oxide via a triplet-triplet annihilation-mediated upconversion mechanism, where the subtle fabrication of rigid polymer shells hosting solvent-phase sensitizer/acceptor pairs is crucial (Figure 5e,f).[77] It should be noted that this is not a strict prototype for regulating the intrinsic high-order correlations in polymers, due to the fact that polymer in this system mainly acts as a host material for encapsulating segments with robust triplet-triplet annihilation. However, this example still presents a model of pursuing high-efficiency polymeric photo-

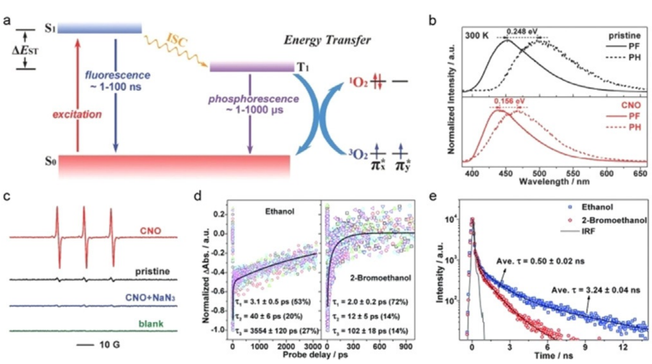

Figure 4. (a) Illustration of the spin-relaxation processes and photocatalytic $^1O_2$ generation in polymeric photocatalysts. (b) Singlet-triplet energy gaps estimated from normalized fluorescence and phosphorescence spectra in pristine and oxidized carbon nitride. (c) Electron spin resonance -trapping tests for detecting $^1O_2$ generation under different catalysts. (d) Transient absorption kinetic traces and (e) time-resolved fluorescence kinetics of polymeric carbon nitride in different solvents. (a–c) Adapted with permission from Ref. [26] Copyright 2016 Wiley. (d–e) Adapted with permission from Ref. [33] Copyright 2017 Royal Society of Chemistry.

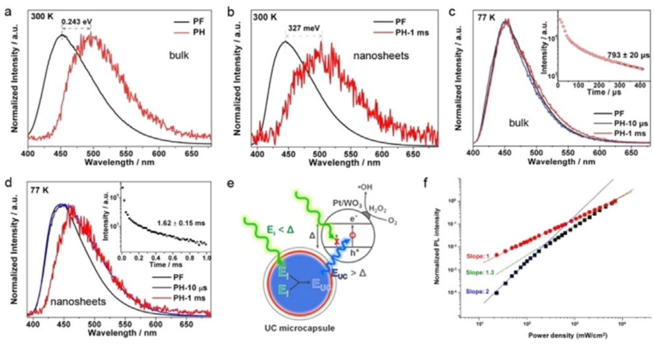

Figure 5. Normalized fluorescence and phosphorescence spectra of (a) bulk polymeric carbon nitride and (b) the corresponding nanosheets under room temperature. Normalized fluorescence and phosphorescence spectra of (c) bulk polymeric carbon nitride and (d) the corresponding nanosheets under 77 K. (e) Schematic illustration of triplet-triplet annihilation-induced sub-band-gap photocatalysis in polymeric upconvertion-microcapsule/platinum-loaded tungsten oxide, and (f) normalized emission intensities of phosphorescence and upconverted fluorescence under different incident-laser power densities in the system. (a–d) Adapted with permission from Ref. [33] 2017 Royal Society of Chemistry. (e–f) Adapted with permission from Ref. [77] Copyright 2012 American Chemical Society.

catalysts by utilizing high-order correlations. In terms of high-order correlations, the high exciton-density thresholds suggest that the efficiencies for utilizing high-order correlations under solar radiation (that is, $100\,\mathrm{mW\,cm^{-2}}$, AM 1.5 G) might be relatively low. To this issue, several strategies including heavy-atom incorporation, dimension reduce, surface plasmon introduction, and core–shell structure construction have been proposed for achieving sunlight-driven high-order correlations.[33,80–82] All these exciting results motivate us to pursue high-efficiency high-order correlations utilization in polymeric photocatalysts in future.

According to the above summary, it is clear that structural factors like heterojunction, heteratom, and dimensionality, which are usually deemed to be effective for regulating band structures and charge-carrier properties, can also implement influences on the excitonic behaviors. Such simultaneous changes in both the excitonic and charge-carrier aspects undoubtedly complicate the estimation of the contribution of excitonic regulation toward and photocatalytic performances. Accordingly, re-examining the intrinsic mechanisms of the optimization strategies, from both excitonic and charge-carrier perspectives, is necessary when dealing with polymer-based photocatalysis.

5. Summary and Outlook

In summary, we here focus on the influences of excitonic effects on polymer-based photocatalysis. Mediated by Coulomb interactions between photoinduced electrons and holes, excitonic effects are pivotal in the photoexcitation processes of polymers, whereas their importance has not been fully realized in the field of polymer-based photocatalysis. In the Minireview, we describe the excitonic aspect of photoexcitation processes in polymeric photocatalysts, where the relationships between the excitonic and charge-carrier aspects are highlighted. In detail, the excitonic and charge-carrier aspects jointly constitute the photoexcitation processes of polymeric photocatalysts, they form a unity of opposites; there are distinct differences between exciton and charge-carrier, such as energy, charge, spin configuration, and high-order correlations, which determine photoexcitation dynamics and hence photocatalytic behaviors therein. According to the understanding of the excitonic aspect in polymeric semiconductors, we further outline the excitonic influences on photocatalysis: the excitonic aspect establishes an alternative mechanism (that is, exciton-based energy transfer) for realizing energy redistribution of the photocatalytic system, beyond the traditional carrier-based charge transfer; the competition between the excitonic and charge-carrier aspects as well as high-order correlations between photoinduced species determine the efficiency of photocatalytic energy utilization in the system. To go further, on the basis of the above comprehension of the excitonic aspect in polymeric photocatalysts, a series of strategies has been proposed for optimizing the excitonic behaviors in polymeric photocatalysts. For instance, order-disorder engineering, heterojunctions, and molecular donor-acceptor blends have been demonstrated to be effective in promoting exciton dissociation; the intrinsic and extrinsic incorporations of heteroatoms are feasible for the spin-relaxation regulation; dimensionality control and heterojunction construction have been proposed for gaining high-efficiency utilization of exciton-exciton annihilation.

Although excitonic effects are of significance in polymeric photocatalysts, the relationships between the excitonic properties and photocatalytic behaviors are far from being sorted out. Therefore, as compared with the well-established charge-carrier one, the excitonic aspect in polymeric photocatalyst deserves more attention in further, and there are still many challenges need to be overcome:

i) It remains difficult to establish accurate relationships between structural factors and excitonic processes in polymeric photocatalysts. Polymeric photocatalysts have long been controversial for their disordered, inhomogeneous structures. In contrast to the rigid structures of inorganic crystals, the much more flexible structures of polymers make it difficult to selectively introduce certain structural factors. Therefore, achieving the synthesis and characterization of well-defined polymers can pave the way for gaining insights into relationships between structural factors and excitonic behaviors. Recent progress in the corresponding fields would be helpful: synthetic methods such as seeded growth,[59] grafting-from strategy,[83] and blocking-cyclization technique[84] exhibit remarkable advantage in precisely constructing well-defined polymers.; techniques like synchrotron radiation-based spectroscopies[67] and electrostatic force microscopy[85] present feasibilities in characterizing the fine structures and electronic properties of polymers. In addition, the adsorption of molecular substrates on the surface of photocatalysts is crucial in exciton-based energy transfer,[62,86] where coordination geometries and reactive intermediates would determine the efficiency and selectivity. To these issues, monitoring the adsorption, activation and desorption of molecular substrates on the photocatalysts surfaces is quite necessary. For instance, reactive sites and coordination geometries can be monitored by in situ FT-IR and XPS,[87,88] while intermediates can be identified by photoionization mass spectrometry.[89] Theoretical study would also be useful to the above problem.[90] Moreover, control tests under certain extrinsic fields are also beneficial for understanding the excitonic aspect in polymeric photocatalysts. For instance, steady-state and time-resolved photoluminescence tests under magnetic field can be used to identify the energy levels and kinetics of excitons with different spin multiplicities.[91] Electric-field-dependent tests can be used for investigating the propertied of charged excitons (or trions) and dark excitons.[92] Space-resolved spectroscopic analyses would be useful for visually identifying the excitonic properties like transport and quenching in polymer-based materials.[93]
ii) Another challenge lies in theoretical simulations. Theoretical simulations have been widely employed in photocatalytic researches, showing great potential in qualitatively describing the relevant electronic and optical properties.[94–96] However, excitonic effects have not been considered in most studies, which usually results in large

deviations. For instance, first-principles density-functional theory (DFT) is typically used to estimate light absorption of photocatalysts (that is, determined by energy gaps between valance and conduction bands). As for polymeric photocatalysts, excitonic effects dominate light absorption, and excitonic absorption leads to smaller optical band gap.[40] Similarly, DFT simulations on transport and photoluminescence properties would be inaccurate once excitonic effects are taken into account. DFT is effective in calculating ground-state structures, whereas is insufficient in dealing with excited-state problems. To this issue, time-dependent density functional theory[43,97] and correlated methods like the GW approximation and the Bethe-Salpeter equation approach[36,98] can be alternatives for determining the electronic and optical properties of polymeric photocatalysts.

iii) The challenge remains in extending the scope of photocatalytic reactions undergoing exciton-based energy transfer pathway. Although excitonic effects dominate photoexcitation processes of polymeric photocatalysts, quite limited reactions, like $^1O_2$ generation and [2+2] styrene cycloadditions directly undergoes exciton-based energy transfer pathway.[99,100] A major reason is that co-existing carrier-based charge transfer presents an important pathway limiting the efficiency and selectivity of exciton-based energy transfer. Therefore, regulating the equilibrium between the two transfers is meaningful for gaining more extensive applications undergoing exciton-based energy transfer. In this respect, it is rational to extend the scope of exciton-based photocatalytic reactions by controlling the electronic coupling between molecular substrates and polymeric photocatalysts.[101] Structural modifications (like element doping, vacancy engineering, co-polymerization, etc.) and co-catalyst designing will be also favorable for regulating the photoexcitation processes and reactive sites in polymeric photocatalysts.

Overall, a crucial fundament should be borne in mind when dealing with polymer-based photocatalysis: the excitonic aspect serves as a supplement to the traditional charge-carrier aspect, and makes a significant contribution to the photocatalytic behaviors of polymers. We believe that the uncovered excitonic aspect in polymeric photocatalysts will open up rich possibilities for polymer-based photocatalytic solar energy utilization.

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Conflict of interest

The authors declare no conflict of interest.

[1] M. A. Fox, M. T. Dulay, *Chem. Rev.* **1993**, *93*, 341.
[2] M. R. Hoffmann, S. T. Martin, W. Y. Choi, D. W. Bahnemann, *Chem. Rev.* **1995**, *95*, 69.
[3] D. M. Schultz, T. P. Yoon, *Science* **2014**, *343*, 1239176.
[4] K. L. Skubi, T. R. Blum, T. P. Yoon, *Chem. Rev.* **2016**, *116*, 10035.
[5] Y. Wang, A. Vogel, M. Sachs, R. S. Sprick, L. Wilbraham, S. J. A. Moniz, R. Godin, M. A. Zwijnenburg, J. R. Durrant, A. I. Cooper, J. Tang, *Nat. Energy* **2019**, *4*, 746.
[6] Y. Wang, X. Wang, M. Antonietti, *Angew. Chem. Int. Ed.* **2012**, *51*, 68; *Angew. Chem.* **2012**, *124*, 70.
[7] G. Zhang, Z. A. Lan, X. Wang, *Angew. Chem. Int. Ed.* **2016**, *55*, 15712; *Angew. Chem.* **2016**, *128*, 15940.
[8] Y. Wang, D. Astruc, A. S. Abd-El-Aziz, *Chem. Soc. Rev.* **2019**, *48*, 558.
[9] W. J. Ong, L. L. Tan, Y. H. Ng, S. T. Yong, S. P. Chai, *Chem. Rev.* **2016**, *116*, 7159.
[10] S. Cao, J. Low, J. Yu, M. Jaroniec, *Adv. Mater.* **2015**, *27*, 2150.
[11] C. Dai, B. Liu, *Energy Environ. Sci.* **2020**, *13*, 24.
[12] M. Z. Rahman, K. Davey, C. B. Mullins, *Adv. Sci.* **2018**, *5*, 1800820.
[13] R. S. Sprick, J.-X. Jiang, B. Bonillo, S. Ren, T. Ratvijitvech, P. Guiglion, M. A. Zwijnenburg, D. J. Adams, A. I. Cooper, *J. Am. Chem. Soc.* **2015**, *137*, 3265.
[14] K. Zhang, D. Kopetzki, P. H. Seeberger, M. Antonietti, F. Vilela, *Angew. Chem. Int. Ed.* **2013**, *52*, 1432; *Angew. Chem.* **2013**, *125*, 1472.
[15] S. Chu, Y. Wang, Y. Guo, P. Zhou, H. Yu, L. L. Luo, F. Kong, Z. G. Zou, *J. Mater. Chem.* **2012**, *22*, 15519.
[16] R. S. Sprick, B. Bonillo, R. Clowes, P. Guiglion, N. J. Brownbill, B. J. Slater, F. Blanc, M. A. Zwijnenburg, D. J. Adams, A. I. Cooper, *Angew. Chem. Int. Ed.* **2016**, *55*, 1792; *Angew. Chem.* **2016**, *128*, 1824.
[17] K. Schwinghammer, S. Hug, M. B. Mesch, J. Senker, B. V. Lotsch, *Energy Environ. Sci.* **2015**, *8*, 3345.
[18] E. Ay, Z. Raad, O. Dautel, F. Dumur, G. Wantz, D. Gigmes, J. P. Fouassier, J. Lalevée, *Macromolecules* **2016**, *49*, 2124.
[19] C.-Y. Hsu, K.-S. Chang, *J. Phys. Chem. C* **2018**, *122*, 3506.
[20] T. Banerjee, K. Gottschling, G. Savasci, C. Ochsenfeld, B. V. Lotsch, *ACS Energy Lett.* **2018**, *3*, 400.
[21] P. Niu, L. C. Yin, Y. Q. Yang, G. Liu, H. M. Cheng, *Adv. Mater.* **2014**, *26*, 8046.
[22] G. Zhang, M. Zhang, X. Ye, X. Qiu, S. Lin, X. Wang, *Adv. Mater.* **2014**, *26*, 805.
[23] S. Ghosh, N. A. Kouame, L. Ramos, S. Remita, A. Dazzi, A. Deniset-Besseau, P. Beaunier, F. Goubard, P. H. Aubert, H. Remita, *Nat. Mater.* **2015**, *14*, 505.
[24] I. Ghosh, J. Khamrai, A. Savateev, N. Shlapakov, M. Antonietti, B. König, *Science* **2019**, *365*, 360.
[25] C. Li, Y. Du, D. Wang, S. Yin, W. Tu, Z. Chen, M. Kraft, G. Chen, R. Xu, *Adv. Funct. Mater.* **2017**, *27*, 1604328.
[26] H. Wang, S. Jiang, S. Chen, D. Li, X. Zhang, W. Shao, X. Sun, J. Xie, Z. Zhao, Q. Zhang, Y. Tian, Y. Xie, *Adv. Mater.* **2016**, *28*, 6940.
[27] X. Zhang, X. Xie, H. Wang, J. Zhang, B. Pan, Y. Xie, *J. Am. Chem. Soc.* **2013**, *135*, 18.

[28] S. Yang, Y. Gong, J. Zhang, L. Zhan, L. Ma, Z. Fang, R. Vajtai, X. Wang, P. M. Ajayan, *Adv. Mater.* **2013**, *25*, 2452.
[29] T. Mueller, E. Malic, *NPJ 2D Mater. Appl.* **2018**, *2*, 29.
[30] Q. Zhang, B. Li, S. Huang, H. Nomura, H. Tanaka, C. Adachi, *Nat. Photonics* **2014**, *8*, 326.
[31] M. Rohlfing, S. G. Louie, *Phys. Rev. Lett.* **1999**, *82*, 1959.
[32] S. Brazovskii, N. Kirova, *Chem. Soc. Rev.* **2010**, *39*, 2453.
[33] H. Wang, S. Jiang, S. Chen, X. Zhang, W. Shao, X. Sun, Z. Zhao, Q. Zhang, Y. Luo, Y. Xie, *Chem. Sci.* **2017**, *8*, 4087.
[34] Y. Xu, S.-P. Gao, *Int. J. Hydrogen Energy* **2012**, *37*, 11072.
[35] S. Melissen, T. Le Bahers, S. N. Steinmann, P. Sautet, *J. Phys. Chem. C* **2015**, *119*, 25188.
[36] W. Wei, T. Jacob, *Phys. Rev. B* **2013**, *87*, 085202.
[37] C. Merschjann, T. Tyborski, S. Orthmann, F. Yang, K. Schwarzburg, M. Lublow, M. C. Lux-Steiner, T. Schedel-Niedrig, *Phys. Rev. B* **2013**, *87*, 205204.
[38] S. Kraner, R. Scholz, C. Koerner, K. Leo, *J. Phys. Chem. C* **2015**, *119*, 22820.
[39] B. Carsten, J. M. Szarko, L. Lu, H. J. Son, F. He, Y. Y. Botros, L. X. Chen, L. Yu, *Macromolecules* **2012**, *45*, 6390.
[40] L. Benatto, M. Koehler, *J. Phys. Chem. C* **2019**, *123*, 6395.
[41] K. L. Corp, C. W. Schlenker, *J. Am. Chem. Soc.* **2017**, *139*, 7904.
[42] Z. Chen, Q. Zhang, Y. Luo, *Angew. Chem. Int. Ed.* **2018**, *57*, 5320; *Angew. Chem.* **2018**, *130*, 5418.
[43] P. Guiglion, C. Butchosa, M. A. Zwijnenburg, *Macromol. Chem. Phys.* **2016**, *217*, 344.
[44] R. Chen, J. Zhu, H. An, F. Fan, C. Li, *Faraday Discuss.* **2017**, *198*, 473.
[45] S. De Rinaldis, I. D'Amico, E. Biolatti, R. Rinaldi, R. Cingolani, F. Rossi, *Phys. Rev. B* **2002**, *65*, 081309(R).
[46] S. Abe, J. Yu, W. P. Su, *Phys. Rev. B* **1992**, *45*, 8264.
[47] D. M. E. Freeman, A. J. Musser, J. M. Frost, H. L. Stern, A. K. Forster, K. J. Fallon, A. G. Rapidis, F. Cacialli, I. McCulloch, T. M. Clarke, R. H. Friend, H. Bronstein, *J. Am. Chem. Soc.* **2017**, *139*, 11073.
[48] E. J. W. List, U. Scherf, K. Müllen, W. Graupner, C. H. Kim, J. Shinar, *Phys. Rev. B* **2002**, *66*, 235203.
[49] F. Steiner, J. Vogelsang, J. M. Lupton, *Phys. Rev. Lett.* **2014**, *112*, 137402.
[50] I. B. Martini, A. D. Smith, B. J. Schwartz, *Phys. Rev. B* **2004**, *69*, 035204.
[51] W. K. Bae, L. A. Padilha, Y.-S. Park, H. McDaniel, I. Robel, J. M. Pietryga, V. I. Klimov, *ACS Nano* **2013**, *7*, 3411.
[52] W. K. Bae, Y. S. Park, J. Lim, D. Lee, L. A. Padilha, H. McDaniel, I. Robel, C. Lee, J. M. Pietryga, V. I. Klimov, *Nat. Commun.* **2013**, *4*, 2661.
[53] T. Nguyen, I. B. Martini, J. Liu, B. J. Schwartz, *J. Phys. Chem. B* **2000**, *104*, 237.
[54] I. A. Howard, J. M. Hodgkiss, X. Zhang, K. R. Kirov, H. A. Bronstein, C. K. Williams, R. H. Friend, S. Westenhoff, N. C. Greenham, *J. Am. Chem. Soc.* **2010**, *132*, 328.
[55] R. Sato, H. Kitoh-Nishioka, K. Kamada, T. Mizokuro, K. Kobayashi, Y. Shigeta, *J. Phys. Chem. Lett.* **2018**, *9*, 6638.
[56] S. M. King, D. Dai, C. Rothe, A. P. Monkman, *Phys. Rev. B* **2007**, *76*, 085204.
[57] P. E. Shaw, A. J. Lewis, A. Ruseckas, I. D. W. Samuel, *Proc. SPIE* **2006**, *6334*, 63340G.
[58] R. Tempelaar, T. L. C. Jansen, J. Knoester, *J. Phys. Chem. Lett.* **2017**, *8*, 6113.
[59] X. H. Jin, M. B. Price, J. R. Finnegan, C. E. Boott, J. M. Richter, A. Rao, S. M. Menke, R. H. Friend, G. R. Whittell, I. Manners, *Science* **2018**, *360*, 897.
[60] H. Wang, X. Sun, D. Li, X. Zhang, S. Chen, W. Shao, Y. Tian, Y. Xie, *J. Am. Chem. Soc.* **2017**, *139*, 2468.
[61] Y. Tamai, H. Ohkita, H. Benten, S. Ito, *J. Phys. Chem. Lett.* **2015**, *6*, 3417.
[62] R. Meng, Y. Li, C. Li, K. Gao, S. Yin, L. Wang, *Phys. Chem. Chem. Phys.* **2017**, *19*, 24971.
[63] A. Migani, L. Blancafort, *J. Am. Chem. Soc.* **2016**, *138*, 16165.
[64] F. Paquin, G. Latini, M. Sakowicz, P. L. Karsenti, L. Wang, D. Beljonne, N. Stingelin, C. Silva, *Phys. Rev. Lett.* **2011**, *106*, 197401.
[65] L. Shi, C. K. Lee, A. P. Willard, *ACS Cent. Sci.* **2017**, *3*, 1262.
[66] H. Ou, X. Chen, L. Lin, Y. Fang, X. Wang, *Angew. Chem. Int. Ed.* **2018**, *57*, 8729; *Angew. Chem.* **2018**, *130*, 8865.
[67] Y. Cao, S. Chen, Q. Luo, H. Yan, Y. Lin, W. Liu, L. Cao, J. Lu, J. Yang, T. Yao, S. Wei, *Angew. Chem. Int. Ed.* **2017**, *56*, 12191; *Angew. Chem.* **2017**, *129*, 12359.
[68] Z.-A. Lan, W. Ren, X. Chen, Y. Zhang, X. Wang, *Appl. Catal. B* **2019**, *245*, 596.
[69] Z. A. Lan, G. Zhang, X. Chen, Y. Zhang, K. A. I. Zhang, X. Wang, *Angew. Chem. Int. Ed.* **2019**, *58*, 10236; *Angew. Chem.* **2019**, *131*, 10342.
[70] J. Kosco, M. Bidwell, H. Cha, T. Martin, C. T. Howells, M. Sachs, D. H. Anjum, S. Gonzalez Lopez, L. Y. Zou, A. Wadsworth, W. M. Zhang, L. S. Zhang, J. Tellam, R. Sougrat, F. Laquai, D. M. DeLongchamp, J. R. Durrant, I. McCulloch, *Nat. Mater.* **2020**, *19*, 559.
[71] J. Ran, W. Guo, H. Wang, B. Zhu, J. Yu, S.-Z. Qiao, *Adv. Mater.* **2018**, *30*, 1800128.
[72] M. Z. Rahman, J. Ran, Y. Tang, M. Jaroniec, S.-Z. Qiao, *J. Mater. Chem. A* **2016**, *4*, 2445.
[73] M. Z. Rahman, K. Davey, S.-Z. Qiao, *J. Mater. Chem. A* **2018**, *6*, 1305.
[74] M. Sachs, R. S. Sprick, D. Pearce, S. A. J. Hillman, A. Monti, A. A. Y. Guilbert, N. J. Brownbill, S. Dimitrov, X. Shi, F. Blanc, M. A. Zwijnenburg, J. Nelson, J. R. Durrant, A. I. Cooper, *Nat. Commun.* **2018**, *9*, 4968.
[75] X. Wang, L. Chen, S. Y. Chong, M. A. Little, Y. Wu, W.-H. Zhu, R. Clowes, Y. Yan, M. A. Zwijnenburg, R. S. Sprick, A. I. Cooper, *Nat. Chem.* **2018**, *10*, 1180.
[76] C. Li, Y. Wang, C. Li, S. Xu, X. Hou, P. Wu, *ACS Appl. Mater. Interfaces* **2019**, *11*, 20770.
[77] J. H. Kim, J. H. Kim, *J. Am. Chem. Soc.* **2012**, *134*, 17478.
[78] O. S. Kwon, J. H. Kim, J. K. Cho, J. H. Kim, *ACS Appl. Mater. Interfaces* **2015**, *7*, 318.
[79] C. Ye, J. Wang, X. Wang, P. Ding, Z. Liang, X. Tao, *Phys. Chem. Chem. Phys.* **2016**, *18*, 3430.
[80] Y. L. Yu, Y. F. Yu, C. Xu, A. Barrette, K. Gundogdu, L. Y. Cao, *Phys. Rev. B* **2016**, *93*, 201111(R).
[81] M. A. Mahmoud, A. J. Poncheri, R. L. Phillips, M. A. El-Sayed, *J. Am. Chem. Soc.* **2010**, *132*, 2633.
[82] J. H. Kang, E. Reichmanis, *Angew. Chem. Int. Ed.* **2012**, *51*, 11841; *Angew. Chem.* **2012**, *124*, 12011.
[83] B. Xu, C. Feng, X. Huang, *Nat. Commun.* **2017**, *8*, 333.
[84] J. Chen, H. Li, H. Zhang, X. Liao, H. Han, L. Zhang, R. Sun, M. Xie, *Nat. Commun.* **2018**, *9*, 5310.
[85] J. S. Harrison, D. A. Waldow, P. A. Cox, R. Giridharagopal, M. Adams, V. Richmond, S. Modahl, M. Longstaff, R. Zhuravlev, D. S. Ginger, *ACS Nano* **2019**, *13*, 536.
[86] C. Mongin, S. Garakyaraghi, N. Razgoniaeva, M. Zamkov, F. N. Castellano, *Science* **2016**, *351*, 369.
[87] R. Long, Y. Li, Y. Liu, S. M. Chen, X. S. Zheng, C. Gao, C. H. He, N. S. Chen, Z. M. Qi, L. Song, J. Jiang, J. F. Zhu, Y. J. Xiong, *J. Am. Chem. Soc.* **2017**, *139*, 4486.
[88] X. Li, Y. Sun, J. Xu, Y. Shao, J. Wu, X. Xu, Y. Pan, H. Ju, J. Zhu, Y. Xie, *Nat. Energy* **2019**, *4*, 690.
[89] Y. Zhu, X. Chen, Y. Wang, W. Wen, Y. Wang, J. Yang, Z. Zhou, L. Zhang, Y. Pan, F. Qi, *Energy Fuels* **2016**, *30*, 1598.
[90] P. Huo, D. F. Coker, *J. Phys. Chem. Lett.* **2011**, *2*, 825.
[91] A. Köhler, H. Bassler, *Mater. Sci. Eng. R* **2009**, *66*, 71.

[92] Z. Li, T. Wang, Z. Lu, M. Khatoniar, Z. Lian, Y. Meng, M. Blei, T. Taniguchi, K. Watanabe, S. A. McGill, S. Tongay, V. M. Menon, D. Smirnov, S.-F. Shi, *Nano Lett.* **2019**, *19*, 6886.
[93] J. J. Crochet, J. G. Duque, J. H. Werner, S. K. Doorn, *Nat. Nanotechnol.* **2012**, *7*, 126.
[94] Y. Li, Y.-L. Li, B. Sac, R. Ahujad, *Catal. Sci. Technol.* **2017**, *7*, 545.
[95] T. Su, Q. Shao, Z. Qin, Z. Guo, Z. Wu, *ACS Catal.* **2018**, *8*, 2253.
[96] Y. Bai, L. Wilbraham, B. J. Slater, M. A. Zwijnenburg, R. S. Sprick, A. I. Cooper, *J. Am. Chem. Soc.* **2019**, *141*, 9063.
[97] S. A. Mewes, F. Plasser, A. Dreuw, *J. Phys. Chem. Lett.* **2017**, *8*, 1205.
[98] B. Bagheri, B. Baumeier, M. Karttunen, *Phys. Chem. Chem. Phys.* **2016**, *18*, 30297.
[99] M. C. DeRosa, R. J. Crutchley, *Coord. Chem. Rev.* **2002**, *233–234*, 351.
[100] Z. Lu, T. P. Yoon, *Angew. Chem. Int. Ed.* **2012**, *51*, 10329; *Angew. Chem.* **2012**, *124*, 10475.
[101] C.-P. Hsu, *Acc. Chem. Res.* **2009**, *42*, 509.

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